[Def.
A deformation retraction of a space X onto a subspace A is a family
$\{f_t : X \to X \mid t \in I\}$ satisfying $f_0 \equiv id$, $f_1[X] = A$, $f_t|_A \equiv id_A$ for all t,
and $X \times I \to X : (x,t) \mapsto f_t(x)$ is continuous.

More generally, define
[Def.
A Homotopy of a space X into a space Y is a family
$\{f_t : X \to Y \mid t \in I\}$ satisfying the associated map $F : X \times I \to Y : (x,t) \mapsto f_t(x)$ is continuous.

Moreover, given two maps g_0, g_1 are called Homotopic if there is a homotopy.
In such a situation, denote $g_0 \simeq g_1$

Recall that a map $r : X \to X$ satisfying $r[X] = A$ and $r|_A \equiv id_A$ is called retraction.
Thus, we can say that the deformation retraction of a space X onto a subspace A is a homotopy from the identity map of X to a retraction of X onto A.
And, algebraically, the retraction $r : X \to X$ satisfies that $r \circ r = r^2 = r$,
analogous to the projection operator.

[Def.
A map $f : X \to Y$ is called a homotopy equivalence if there exists a map $g : Y \to X$ s.t
$$f \circ g \simeq id_Y \quad \text{and} \quad g \circ f \simeq id_X.$$

If such a map exists, then the spaces X and Y are called homotopy equivalent,
or Homotopic.
[Thm. If X and Y are Homeomorphic, then X and Y are Homotopic.
Proof. Let $f : X \to Y$ be a Homeomorphism. Then,
$$f^{-1} \circ f = id_X \simeq id_X \quad \text{and} \quad f \circ f^{-1} = id_Y \simeq id_Y,$$
So proved. ☐

Contractible

Def.
A space that homotopy equivalent to a point is called Contractible.

Thm. Let X be a Topological space. Then, TFAE:
a). X is Contractible. (Homotopy equivalent to a point)
b). $id_X : X \to X ; x \mapsto x$ is null-homotopic.
c). Given $x \in X$, $C_x \simeq id_X$ where $C_x : X \to X ; z \mapsto x$.

Proof.
Assume a). That is, there exist two maps
$f : X \to \{p\}$ and $g : \{p\} \to X$ such that $f \circ g \simeq id_p$ and $g \circ f \simeq id_X$.

Then, Since $f : X \to \{p\}$ must be a constant mapping $x \mapsto p$,
$g \circ f(x) = g(p)$ for all $x \in X$, so $g \circ f$ is constant, so id_X is null-homotopic.
It proved b).
Assume b). Let $C : X \to X ; z \to a$ be a Constant map such that $C \simeq id_X$, via
$H : X \to I \to X$ such that $H(z,0) = z$ and $H(z,1) = a$.

Let $x \in X$ be fixed. Define a path $P : [0,1] \to X ; t \mapsto H(x,t)$.
Then, $P(0) = H(x,0) = x$ and $P(1) = H(x,1) = a$.
Now, $F : X \times I \to X ; (x,t) \mapsto P(t)$ is the Homotopy to $C \simeq C_x$. Now, by transitivity.
$C \simeq id_X$ and $C \simeq C_x$ implies that $C_x \simeq id_X$. It proved c).
Assume c). Let $f : X \to \{p\} ; x \mapsto p$ be a map and let $x \in X$ be given.
Then, a map $g : \{p\} \to X ; p \mapsto x$ satisfies

$$f \circ g = id_p \text{ and } g \circ f = C_x.$$

By c), $C_x \simeq id_X$, so $g \circ f \simeq id_X$. So proved ∎

Corollary. Contractible space is Path-Connected.
Proof. Suppose that X is Contractible space. Let $x, y \in X$ be given.
Then, $C_x \simeq id_X \simeq C_y$, so there exists a Homotopy

$$F : X \times I \to X \text{ such that } \begin{cases} F(z,0) = C_x(z) = x & \text{for all } z \in X \\ F(z,1) = C_y(z) = y & \text{for all } z \in X. \end{cases}$$

Now, $P : I \to X ; t \mapsto F(x,t)$ is a path from x to y. ∎